

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.28462	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.05894	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1581.8565	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	1.0	0.073643	0.073643	193.70492	2	0	0	0.945921	14.345
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	191.07752	1	0	0	0.982184	13.5
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.070733	0.070733	1358.68936	2	0	0	0.860424	14.775

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.28462.
- Accepted epoch count: 0.

- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.05894.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1581.8565.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.

- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.069472.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073643.
- Held-out runtime ms: 193.70492.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002874.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.099117, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.096475, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.067432, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.047481, errors=[]

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 191.07752.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.065731.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070733.
- Held-out runtime ms: 1358.68936.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.000649.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.096106, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.090814, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.07126, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.049094, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.046393, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility Rate	Heldout Feasible Gap	Heldout Runtime (ms)	Robustness Dispersion	Notes
baseline_nearest_neighbor_constructive	1.0	0.2734	42.3	0.0028	Fastest baseline, moderate gap
baseline_clarke_wright_savings	1.0	0.0738	77.1	0.0078	Good gap, small runtime
baseline_regret_insertion_local_search	1.0	0.4664	1581.9	0.0869	Largest gap, longest runtime
direct_generate_plus_on_e_repair	1.0	0.0736	193.7	0.0029	Learned direct synthesis; gap comparable to Clarke-Wright, moderate runtime
budget_matched_no_replay	0.0	0.0*	191.1	0.0	No feasibility on heldout, very poor performance
replay_solver_evolution	1.0	**0.0707**	1358.7	**0.00065**	Best heldout gap and robustness, high runtime but feasible

* Penalized gap of 3.0 indicates infeasibility.

Key Takeaways

- **Direct synthesis** (direct_generate_plus_one_repair) achieved feasibility and a heldout gap comparable to best fixed baselines (Clarke-Wright), with moderate runtime and good robustness.
- **Replay-aware iterative search** (replay_solver_evolution) showed statistically the best heldout objective gap and the highest robustness, beating both fixed baselines and direct synthesis albeit with longer runtime.
- The **budget-matched no-replay control** completely failed on feasibility, indicating the benefit of replay in iterative search.
- Among fixed baselines, Clarke-Wright Savings achieves a strong balance of feasibility, objective gap, and runtime.

Conclusions

- No learned condition clearly outperforms all fixed baselines across all criteria, but **replay solver evolution** is the best in objective gap and robustness, demonstrating benefits of replay.
- Replay-aware iterative search **clearly outperforms its budget-matched no-replay counterpart**, which fails feasibility entirely.
- Direct generation plus repair is a viable learned method, matching fixed baseline gaps with reasonable runtimes.

This suggests **replay and evolutionary iterative improvements provide clear advantages** in Phase-9 CVRP closeout over fixed baselines and non-replay learned approaches.